

Homework 3 ECEN 4293 Numerical Methods in Python for Engineers

Student: Ricardo Landeros Aranda

Instructor: Marcus Mellor

April 3rd 2026

1 Problem 1

1.1 Part a)

Testing the 3-point density interpolation plot from lecture we get the following:

```
For the given array in class when we look for the vandemonde interpolation we get: 0.567625.  
From the following resultant polynomial: 1.027 - 0.001715 x + (1.15e-06) x**2.
```

Figure 1: Vandermonde interpolation used to solve for a point in between given data and its resultant polynomial.

1.2 Part b)

Once the Vandermonde Interpolation was used to calculate the point lying between the 300 and 400 mark in part a) I proceeded to solve for the interpolation of the function $f(x) = \cos(2\pi x)$.

To do this, I simply set a couple of node arrays for the function and fed these into the Vandermonde interpolator building the (x_i, y_i) pairs. Once we ran multiple calculations, when changing the number of interpolation points everytime we graph we appreciate how the polynomial approximates the real function more and more. Especially once we incremented the sample size to 9 or 10 the interpolation result approached the real curve almost in its entirety. In the figures below we can appreciate the reach of the polynomial with interpolation 2, 4, 6, and 12 points respectively. This is shown in Figure 2 below.

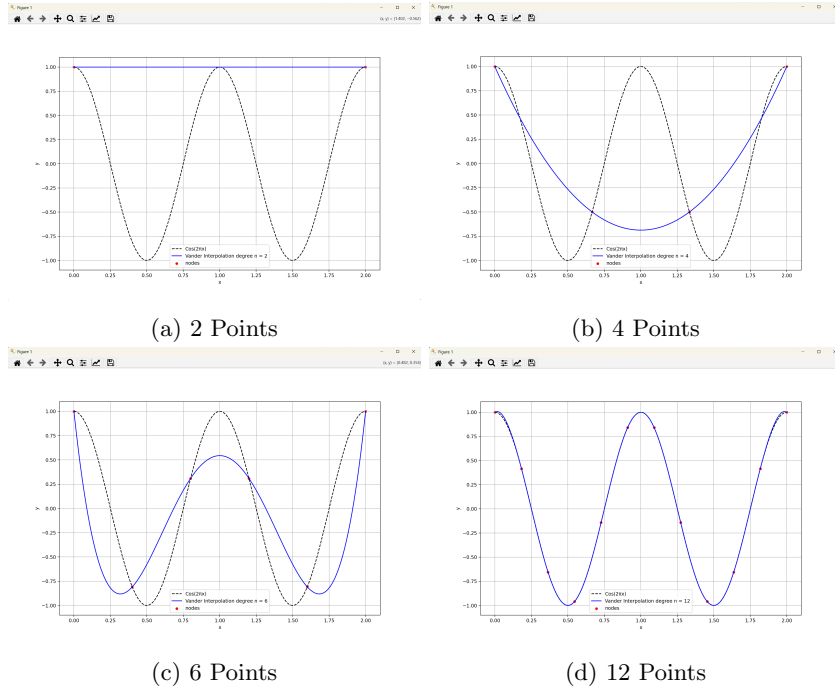


Figure 2: Vandermonde polynomial interpolation: Multiple Interpolation points.

2 Problem 2

2.1 Part b)

For this part, I decided to compare the exact same test used with the Vandermonde interpolation used to approach the function $f(x) = \cos(2\pi x)$. Adding another variable to store the Lagrange polynomial, I ran the same test over the same interpolation points. Surprisingly or unsurprisingly, the interpolations for both Vandermonde and Lagrange are **very** close. The plot resulting from the test is shown below in Figure 3.

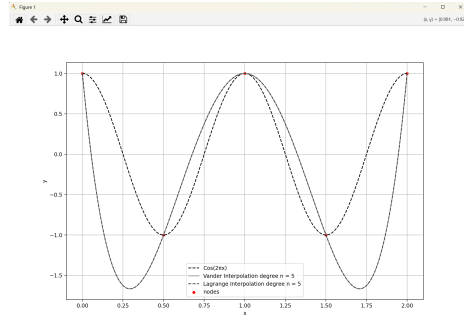


Figure 3: Vandermonde and Lagrange interpolation plots.

3 Problem 3

Lastly, this problem resulted in a fairly trivial execution since I just had to rewrite the table in array form, obtain the polynomial from the interpolation, and finally apply bisection to the polynomial over a range from 4861 to 4340. Below, in Figure 4 shows the resultant polynomial. Which once I used bisection to find the wavelength at an index refraction of 1.520 is 4600.5 .

```
The resultant polynomial is: 29.06715812 - 0.03648898 x + (2.06323342e-05) x**2 -
(6.45218924e-09) x**3 + (1.28438366e-12) x**4 - (1.341884e-16) x**5 +
(8.24979339e-21) x**6 - (2.16289895e-25) x**7
```

Figure 4: Resultant polynomial of applying Lagrange interpolation to the table given of wavelength and refractive index.